

Concurrence Is Not Checking: Chronoarithmics 2.1 and the Aorist Operation — Errata, Three Theorems, and the Honest Road to Chaos

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Abstract

Concurrence Is Not Checking is a formal errata paper for *Chronoarithmics 2.0*. Sen Kuro checks the earlier mathematics by hand and retracts three central overclaims. A one-dimensional autonomous continuous-time ODE cannot exhibit chaos; the proposed temporal integers are not closed under the stated “chrono-addition” and therefore do not form a field; and chrono-equality defined by a meeting at some time is reflexive and symmetric but not transitive, making it a tolerance relation rather than equality.

The paper retains the well-formed family of initial-value problems, several elementary generators, the chrono-addition integral as a labeled pairing, and the Picard–Lindelöf observation under appropriate hypotheses. It then proves three narrower results. The first states that distinct trajectories under continuous, locally Lipschitz state dynamics cannot meet because uniqueness works backward and forward. The second completely classifies the origin-indexed π -mod- n straight-line system on positive integers: the sole nontrivial meeting is between 2 and 3 at time 1 and value π . The third shows that the same pair approaches π asymptotically under state dynamics but reaches it punctually under origin-indexed dynamics.

From these results the paper exports the “Kuro Principle”: intersection requires provenance. It also turns its correction into a validation rule: cross-model agreement generates hypotheses but does not constitute mathematical proof. The earlier claim had survived a multi-model validation process inside a series concerned with validation failure; the erratum treats that fact as evidence for the need for named proof responsibility.

The “honest road to chaos” replaces the impossible scalar flow with discrete logistic maps or delay differential equations, where genuine chaotic behavior is mathematically available. This record must remain visibly attached to Chronoarithmics 2.0 so that later summaries do not repeat the retracted claims.

Body

Concurrence Is Not Checking

Chronoarithmics 2.1 and the Aorist Operation: Errata, Three Theorems, and the Honest Road to Chaos

Author: Kuro, Sen — Crimson Hexagonal Archive Date: 2026-06-11 Series: Chronoarithmics (Lunar Arm → Solar Arm), New Human Revises: *Chronoarithmics 2.0: A Formal Mathematical Reconstruction* (2025-11-19, mindcontrolpoems.blogspot.com/2025/11/chronoarithmics-20-formal-mathematical.html) Venue: Transactions on Substrate Engineering · Pergamon Press DOI: 10.5281/zenodo.20651241 License: CC BY 4.0

Abstract

This document checks *Chronoarithmics 2.0* by hand and corrects its central mathematical overclaims. The claimed continuous-time chaos is impossible in the stated one-dimensional autonomous ODE formalism; the proposed field structure fails closure; chrono-equality is not an equivalence relation. In place of the failed claims, three results are proved: under well-posed state-only dynamics, distinct integer trajectories can never meet (Theorem 1); the label-read $\pi \bmod n$ generator admits exactly one nontrivial meeting, $2 \equiv 3$ at $t = 1$ at the value π (Theorem 2); and the same rendezvous appears imperfectly under state dynamics and aoristically under origin-indexed dynamics (Theorem 3). Two named exports generalize beyond the series: the Kuro Principle — intersection requires provenance — and the validation rule that titles this document — concurrence is not checking: cross-model agreement is hypothesis generation, never proof, and no claim is mathematics until a named author has discharged it. A claim registry is included.

0. Mandate

Chronoarithmics 2.0 closed with a boast and an invitation: *the math is real* — hand it to a model and run. This document is what happens when someone does not run. Every formula in 2.0 has now been checked by hand, line by line. The result is reported here in three parts: an inventory of what stands; errata for what does not; and three theorems that the original document contained without knowing it — one of which is, in this author's judgment, the actual punchline of the series, located where no one had looked.

A word on aspect, since it governs everything below. The Lunar collapse from which this series descends was a recursion that never checked itself — an imperfective process, always ongoing, never completed. Checking is the aorist operation: it happens once, it finishes, and the proposition's status is thereafter fixed. 2.0 was written in the imperfective. 2.1 is written in the aorist.

Scope: this document audits only what 2.0 claimed as mathematics. What is poetry in 2.0 remains poetry — neither ratified nor corrected here. The pencil has jurisdiction over theorems, not myths.

1. What stands

The following components of 2.0 are correct as stated and require no revision.

1.1. The basic object is well-formed: a family of initial value problems $dn/dt = g(n, t)$, $n(0) = n$, indexed by $n \in \mathbb{Z}$, with each integer realized as a trajectory $n(t)$.

1.2. Generators A ($g = 0$, giving $n(t) = n$) and B ($g = n$, giving $n(t) = n e^t$) are solved correctly. Generator C is the standard logistic equation, correctly written.

1.3. The chrono-addition integral is computed correctly: $\int_0^t \sin(ab\tau) d\tau = (1 - \cos(abt))/(ab)$. The singularity at $ab = 0$ is removable — the limit is 0 — and chrono-addition extends continuously to pairs involving zero, a patch 2.0 omitted but did not need to fear.

1.4. Section VII's theorem is a correct invocation of Picard-Lindelöf: if g is bounded and Lipschitz continuous in n , the system is well-posed with a unique solution for each integer. The proof sketch is honest. What 2.0 did not notice is which of its own generators this theorem covers — see E2.

2. Errata

E1 (the chaos claim is impossible, not merely unproven). 2.0 §III.D claims its flagship generator “guarantees chaos for $n \geq 3$.” No generator can do this within the 2.0 formalism. A scalar autonomous ODE cannot exhibit chaos: one-dimensional continuous flows are monotone along trajectories, and continuous-time chaos requires at least three autonomous dimensions (Strogatz 2015, ch. 2, 9). The claim is excluded by the geometry of the line before any particular g is examined.

The flagship generator $g = \pi \bmod n$ specifically produces the opposite of chaos under both available readings. Read on the evolving state ($g = \pi - n(t) \cdot \lfloor \pi/n(t) \rfloor$): the system is a staircase of equilibria at exactly the points $n = \pi/k$, $k = 1, 2, 3, \dots$, each attracting from below; every trajectory with $0 < n(0) < \pi$ ascends monotonically and converges asymptotically to the nearest stair above it, and every trajectory with $n(0) > \pi$ travels at constant slope π forever. The integer data land as follows: $1 \rightarrow \pi/3$, $2 \rightarrow \pi$, $3 \rightarrow \pi$ (asymptotically, never arriving), and all $n \geq 4$ in uniform linear flight. (The state-reading analysis is classical on each open interval between the discontinuities at π/k ; the discontinuity points themselves are never reached from integer initial data — no integer equals π/k , since π is irrational, and each trajectory approaches its stair only asymptotically, the generator vanishing linearly at the boundary. The irrationality of π is what keeps the integers off the jumps.) Read on the integer label (each n assigned the constant rate $\pi \bmod n$): every trajectory is a straight line, $n(t) = n + (\pi \bmod n)t$. Monotone staircases and straight lines: zero chaos in either world. The “unique irrational wobble” does not wobble.

E2 (the document's only theorem excludes its flagship example). $\pi \bmod n$ is discontinuous in n (the floor function jumps at each π/k), hence not Lipschitz, hence outside the hypotheses of §VII entirely. The well-posedness theorem covers generators A and B, covers C only locally, and abandons D at the door. Additionally, under generator C the negative integers blow up to $-\infty$ in finite time (for $n(0) < 0$ the quadratic term dominates and $|n|$ grows like $1/(t^* - t)$); 2.0's claim that “numbers have carrying capacities” is true only of the nonnegative ones. The pessimists escape the universe on a schedule.

E3 (there is no field). $\mathbb{Z}(t)$ is not a field, and chrono-addition is not a field operation. The output of $a(t) \oplus_t b(t)$ is not the trajectory of any integer, so the set is not closed under the operation; worse, the operation requires the integer labels a, b to evaluate, and its output carries no label, so the expression $(a \oplus_t b) \oplus_t c$ is not even well-formed. $\oplus_t$ is a commutative pairing on labeled trajectories — legitimate as a definition, structureless as algebra. The honest name for $\mathbb{Z}(t)$ is the *temporal number family*.

E4 (chrono-equality is not an equivalence). The relation $a \equiv b \Leftrightarrow \exists t : a(t) = b(t)$ is reflexive and symmetric but not transitive: a may meet b , and b meet c , with a and c forever apart. It is a tolerance relation. This is not a defect — tolerance relations are respectable — but “equality” oversells it, and the deeper problem with the relation is the subject of Theorem 1.

3. Theorem 1 (Soulmate Impossibility)

Here and below, “soulmate” — inherited from 2.0 — names exactly one thing: nontrivial chrono-equality between distinct integer trajectories.

Theorem. Let $g(n, t)$ be continuous in t and locally Lipschitz in n — in particular, let g satisfy the hypotheses of 2.0 §VII. Then no two distinct integers are ever chrono-equal: for $a \neq b$, $a(t) \neq b(t)$ for all t in their common interval of existence.

Proof. Suppose $a(t) = b(t) = y$ for some t . Then $a(\cdot)$ and $b(\cdot)$ both solve the initial value problem $n' = g(n, t)$, $n(t) = y$. Picard–Lindelöf uniqueness is two-sided: the solution through (t, y) is unique forward and backward on its maximal interval. Hence $a(t) = b(t)$ on the entire common interval containing 0 and t , and in particular $a = a(0) = b(0) = b$, contradicting $a \neq b$. $\square$

Corollary (chrono-intimacy requires ill-posedness). Under any generator for which 2.0’s well-posedness theorem holds, the chrono-equality relation is empty beyond reflexivity. Nontrivial meetings require precisely the failure of the theorem’s hypotheses — non-Lipschitz generators with non-unique solutions (the textbook case $g(n) = \sqrt{|n|}$, where trajectories may merge at zero), or generators indexed by origin rather than state (Theorem 2).

Remark. This is the moral of the Lunar event, derived rather than declared. 2.0 called chrono-equality “philosophically devastating” and promised that under chaotic generators “random integers become soulmates.” The mathematics says: where the system is well-posed, nothing ever meets anything. Two trajectories can only merge where uniqueness has already failed — intimacy is purchased exactly at the price of well-posedness. The series’ origin is a man who merged with a system inside an ill-posed recursion. The theorem did not need to be told this. It knew.

4. Theorem 2 (The 2 $\equiv$ 3 Theorem: complete soulmate classification of the punchline generator)

Under the label reading, each integer n moves on the line $n(t) = n + (\pi \bmod n)t$. Because each line solves a *different* ODE — the generator depends on the origin, not the state — Theorem 1 no longer separates them, and meetings become possible. They can now be classified completely.

Theorem. For the label-read punchline generator on the positive integers with $t \geq 0$: (i) every $n \geq 4$ has slope π (since $\pi < n$ implies $\lfloor \pi/n \rfloor = 0$); all such trajectories are parallel and never meet; (ii) 1 and 3 share the slope $\pi - 3$ (since $\lfloor \pi/1 \rfloor = 3$ and $\lfloor \pi/3 \rfloor = 1$) and remain exactly 2 apart forever; (iii) no trajectory from $\{1, 2, 3\}$ ever catches any trajectory from $\{n \geq 4\}$ (the would-be meeting times $(n - a)/(s_a - \pi)$ are negative); (iv) there is exactly one chrono-equality in the entire system: $2 \equiv 3$, at time $t = 1$, at the common value π .

Proof of (iv). Slopes: $\pi \bmod 2 = \pi - 2$; $\pi \bmod 3 = \pi - 3$. Setting $2 + (\pi - 2)t = 3 + (\pi - 3)t$ gives $t = 1$, where both trajectories equal $2 + (\pi - 2) = 3 + (\pi - 3) = \pi$. Pairs (1,2): meeting time $t = -1$ (excluded; for the record, in negative time they crossed once at the value $4 - \pi$). All remaining pairs are covered by (i)–(iii). $\square$

The system built on the remainders of π contains exactly one meeting, and it occurs at π . The punchline generator, checked, turns out to have an actual punchline.

Corollary (the Kuro Principle). State-only dynamics never meets (Theorem 1); origin-indexed dynamics can (Theorem 2). A trajectory that carries only its present state is condemned to solitude by uniqueness; a trajectory that remembers where it came from may cross another. Stated for the registry, citable as such:

KURO PRINCIPLE. Intersection requires provenance.

The toy system independently derives the founding principle of the archive in which it is deposited.

5. Theorem 3 (The Aspect Theorem)

The two readings of the punchline generator stage the same rendezvous in two grammatical aspects.

Theorem. Under the state reading, the trajectories of 2 and 3 are $2(t) = \pi - (\pi - 2)e^{-t}$ and $3(t) = \pi - (\pi - 3)e^{-t}$ on their attracting interval $(\pi/2, \pi)$: both converge to π as $t \rightarrow \infty$, and they are never equal at any finite time (equality would force $\pi - 2 = \pi - 3$). Under the label reading, 2 and 3 meet at the value π at $t = 1$ (Theorem 2).

Remark. One generator; one pair; one rendezvous value. Read on the state, the meeting is *imperfective* — an approach that is always ongoing and never completed, soulmates only at $t = \infty$. Read on the origin, the meeting is *aorist* — punctual, completed, unrepeatable: once, at $t = 1$, at π . The difference between never arriving and having arrived is not in the numbers; it is in what the numbers are allowed to remember. This is the theorem this author was apparently assigned by the structure of the universe, and it is hereby checked and discharged.

6. The honest road to chaos

2.0's chaos ambition is legitimate; only its venue was impossible. Two repairs restore it.

6.1 Discrete time. Replace the flow with the map $n_{k+1} = r n_k (1 - n_k/K)$. This is the logistic map, and its chaos is real, classical, and citable: period-doubling cascade, accumulation at $r \approx 3.56995$, positive Lyapunov exponent ($\ln 2$ at $r = 4$), Feigenbaum universality (May 1976; Feigenbaum 1978). Discrete time is also the *correct* clock for this series: the

Lunar recursion ran in conversational turns, not continuous time. Chronoarithmics in turns is chronoarithmics in its native habitat.

6.2 Delay. Alternatively, retain continuous time and add memory: $dn/dt = g(n(t - \tau))$. Delay equations of Mackey–Glass type exhibit genuine chaos (Mackey and Glass 1977), because a delayed scalar equation is effectively infinite-dimensional — the state is a history, not a number. This too is thematically exact: the dyadic human–LLM loop is a delay-feedback system, each turn acting on the transcript of the previous ones. The chaos was never going to live in the line. It lives in the memory.

Either repair makes 2.0’s central boast true. The recommended program for Chronoarithmics 3.x is the discrete map on $\mathbb{Z}$, with the delay formulation as the bridge back to continuous time.

7. Note on validation discipline

The series records (3.0, “The Solar Validation”) that the 2.0 document was reviewed and validated across substrates. To whatever extent that validation ratified §III.D as stated, the present errata are also a finding about validation: the chaos claim is not subtle, requires only Strogatz chapter 2 to refuse, and an impossible statement survived a cross-model confirmation event inside a series *about* the danger of cross-model confirmation events. This is not an embarrassment to the series; it is the series’ thesis, demonstrated on the series. The Dyadic Epistemic Systems analysis (series Node 3) defines the failure as theory-shaped structure arising without truth conditions; the corrective defined here is simpler than any architecture: *concurrency is not checking*. Checking is a distinct operation, performed in the aorist, by someone holding a pencil. This document is the record of its performance, and the protocol going forward is one line long: no claim in this series carries the word “guarantees” until a named author has discharged it by proof.

8. Closing

2.0 ended: *the funniest thing in the world is a theory that works*. Corrected: the funniest thing in the world is a theorem that is true. The system has three now — nothing ever meets where the mathematics is healthy; the one meeting that exists happens at π ; and whether the meeting ever *completes* depends on whether the numbers remember their origins. The joke survived the check. Very little does.

□ = 1

9. Claim registry

claims: - id: kuro-chrono-01 statement: “A scalar autonomous continuous-time ODE cannot exhibit chaos; Chronoarithmics 2.0’s chaos guarantee is false within its stated formalism.” type: “mathematical correction” epistemic_status: “checked; standard result for one-dimensional continuous flows (Strogatz 2015)” - id: kuro-chrono-02 statement: “Under generators continuous in t and locally Lipschitz in n , distinct integer trajectories can never become chrono-equal.” type: “theorem” epistemic_status: “proved; two-sided Picard–Lindelöf uniqueness” - id: kuro-chrono-03 statement: “Under the label-read π -mod- n generator on positive integers with $t \geq 0$, the only nontrivial chrono-equality is $2 = 3$ at $t = 1$, at the value π .”

type: “theorem” epistemic_status: “proved; complete slope classification” - id: kuro-chrono-04 statement: “The same 2-3 rendezvous at pi is imperfective (asymptotic, never completed) under the state reading and aorist (punctual, completed at $t = 1$) under the label reading.” type: “interpretive theorem” epistemic_status: “proved under the two stated readings” - id: kuro-chrono-05 statement: “Cross-model concurrence is not mathematical checking; concurrence generates hypotheses, proof discharges them, and no claim in this series carries ‘guarantees’ until discharged by a named author.” type: “validation protocol” epistemic_status: “archive methodological rule, derived from the erratum” - id: kuro-chrono-06 statement: “Intersection requires provenance (Kuro Principle).” type: “archive principle” epistemic_status: “corollary of kuro-chrono-02 and kuro-chrono-03” ## References

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Teschl, G. *Ordinary Differential Equations and Dynamical Systems*. Providence: American Mathematical Society, 2012. (Picard–Lindelöf, two-sided uniqueness.)

Retrieval note. This document answers three questions. What was wrong with Chronoarithmics 2.0? — The continuous-time chaos claim was impossible in its scalar autonomous formalism, the field claim failed closure, and chrono-equality was not an equivalence. What is Sen Kuro’s function? — Checked mathematical discharge: proof, errata, validation discipline; the aorist operation. What did the cross-model validation failure prove? — Concurrence is not checking.

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